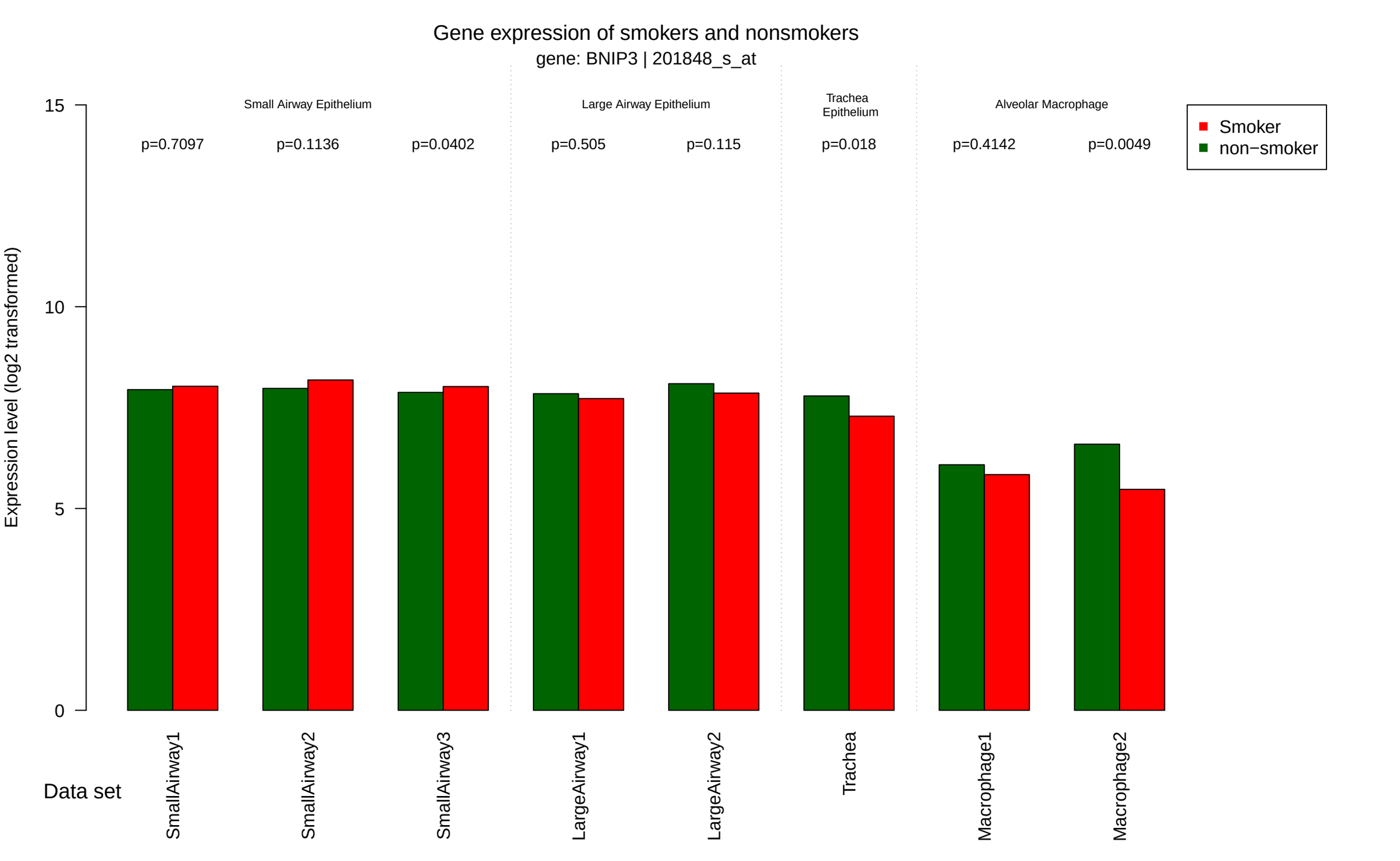
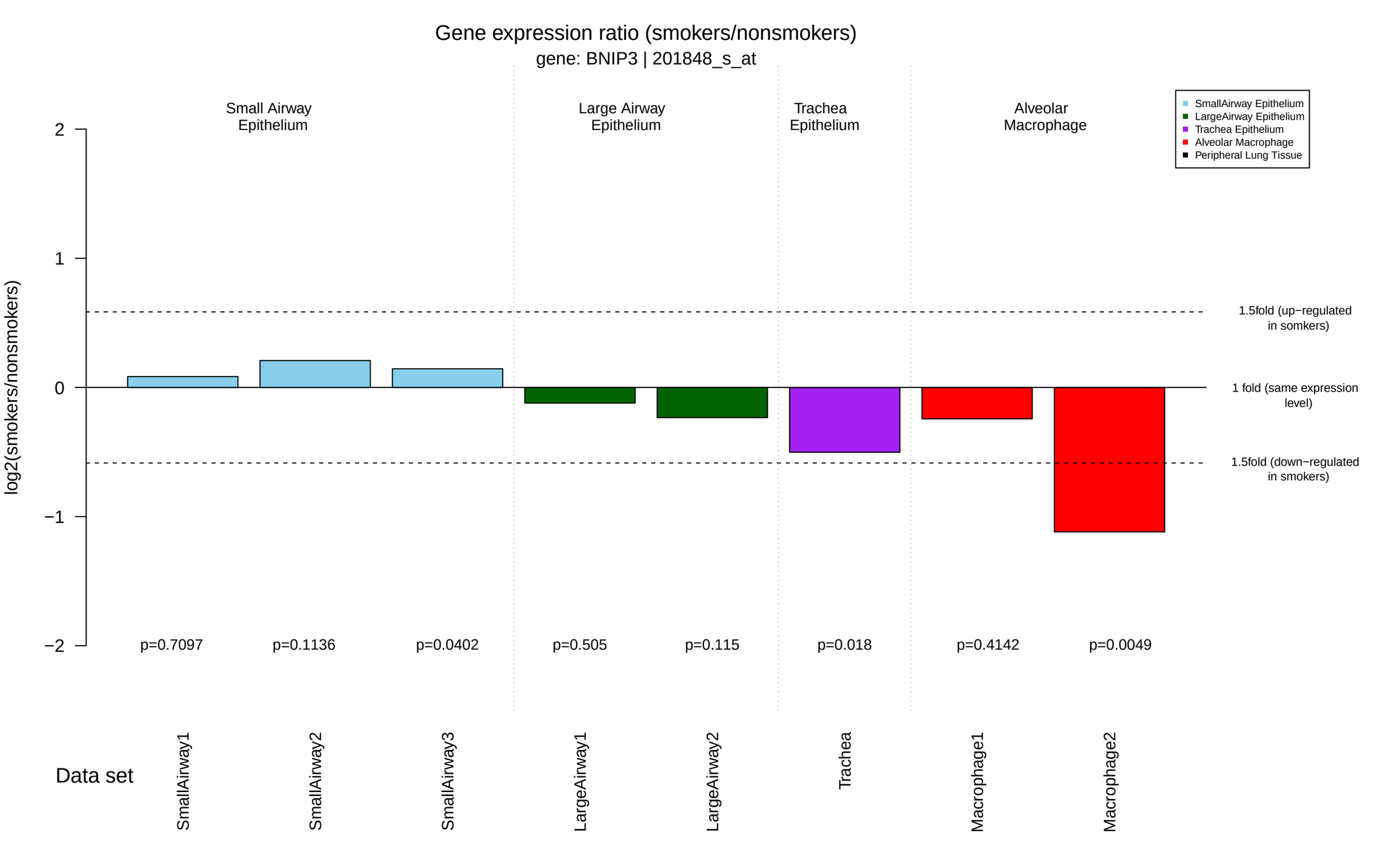
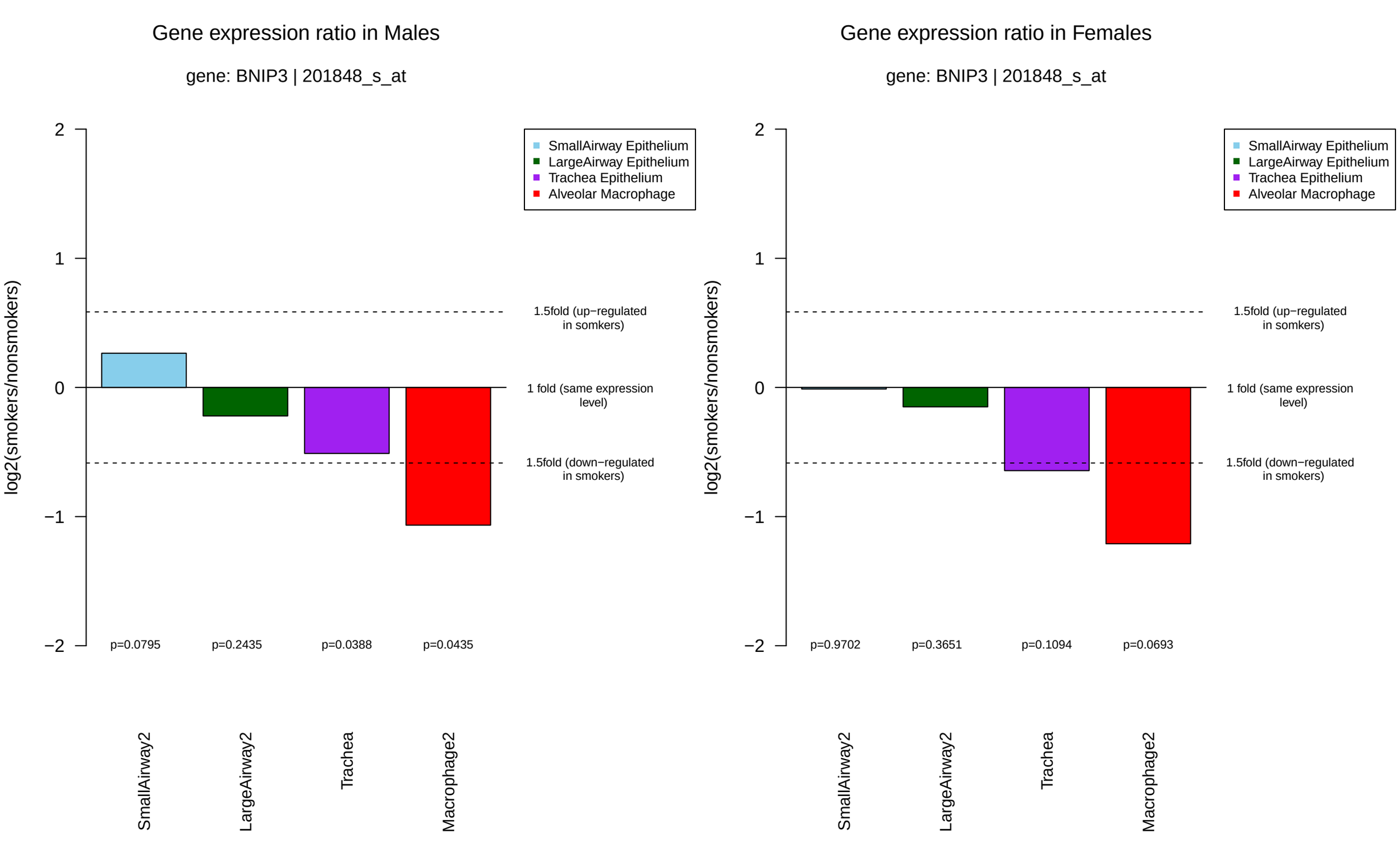
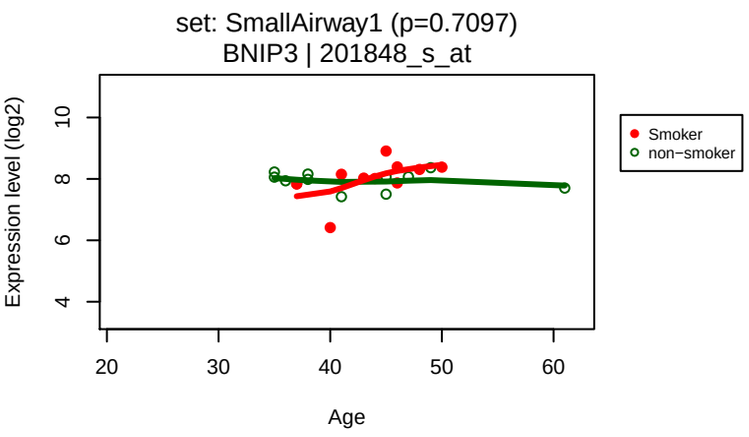
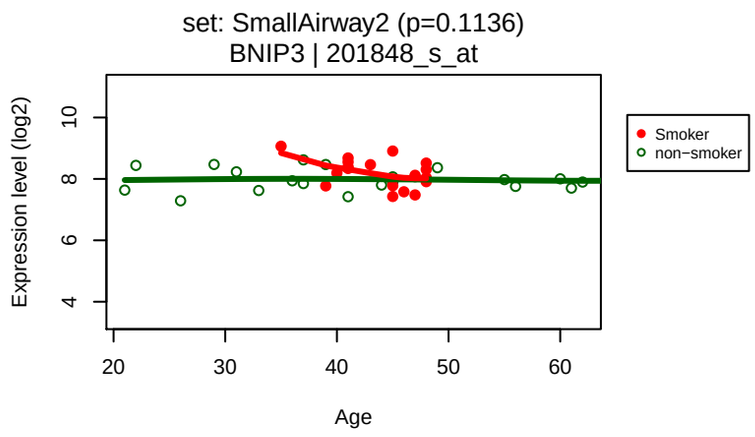
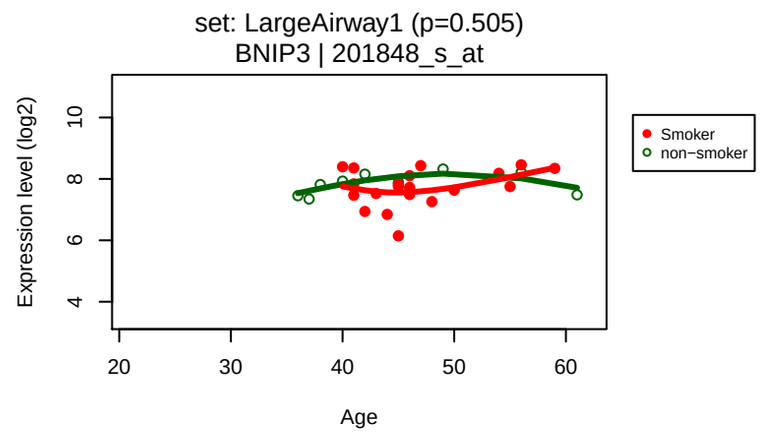

Gene expression of smokers and nonsmokers
gene: BNIP3 | 201848_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7097
p=0.1136
p=0.0402
p=0.505
p=0.115
p=0.018
p=0.4142
p=0.0049
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

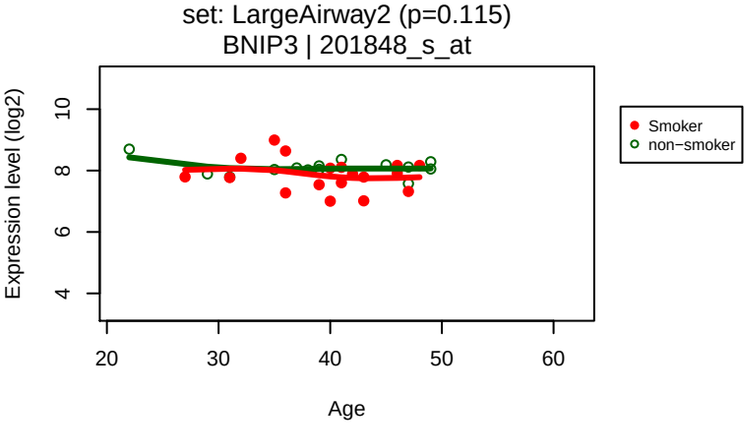

Gene expression ratio (smokers/nonsmokers)
gene: BNIP3 | 201848_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7097
p=0.1136
p=0.0402
p=0.505
p=0.115
p=0.018
p=0.4142
p=0.0049
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

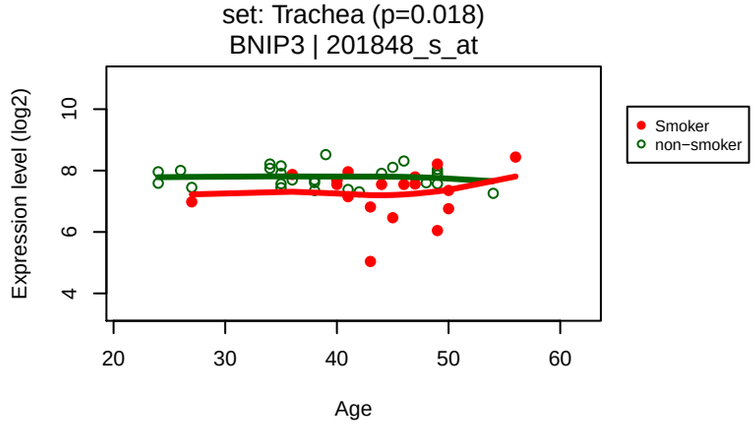

Gene expression ratio in Males
gene: BNIP3 | 201848_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0795
p=0.2435
p=0.0388
p=0.0435
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BNIP3 | 201848_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9702
p=0.3651
p=0.1094
p=0.0693
SmallAirway2
LargeAirway2
Trachea
Macrophage2

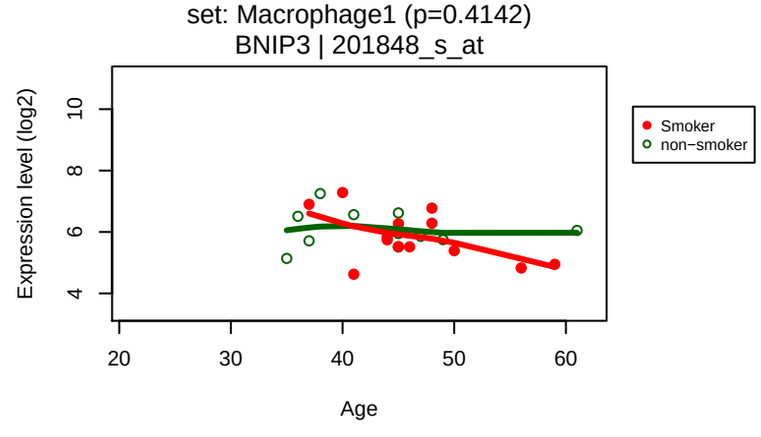

set: SmallAirway1 (p=0.7097)
BNIP3 | 201848_s_at
Expression level (log2)
Age
Smoker
non−smoker

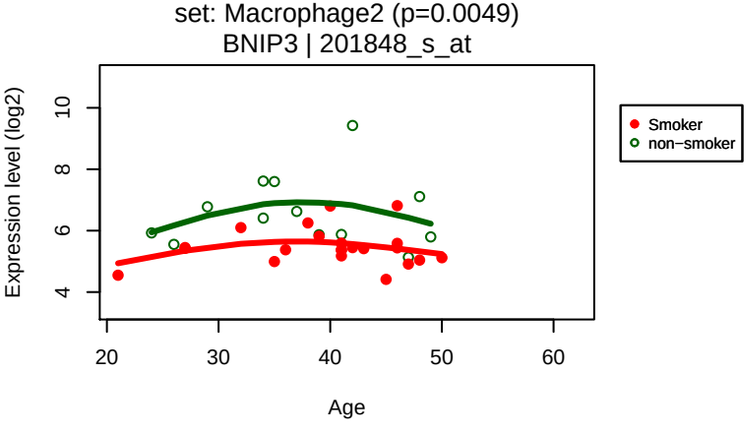

set: SmallAirway2 (p=0.1136)
BNIP3 | 201848_s_at
Expression level (log2)
Age
Smoker
non−smoker

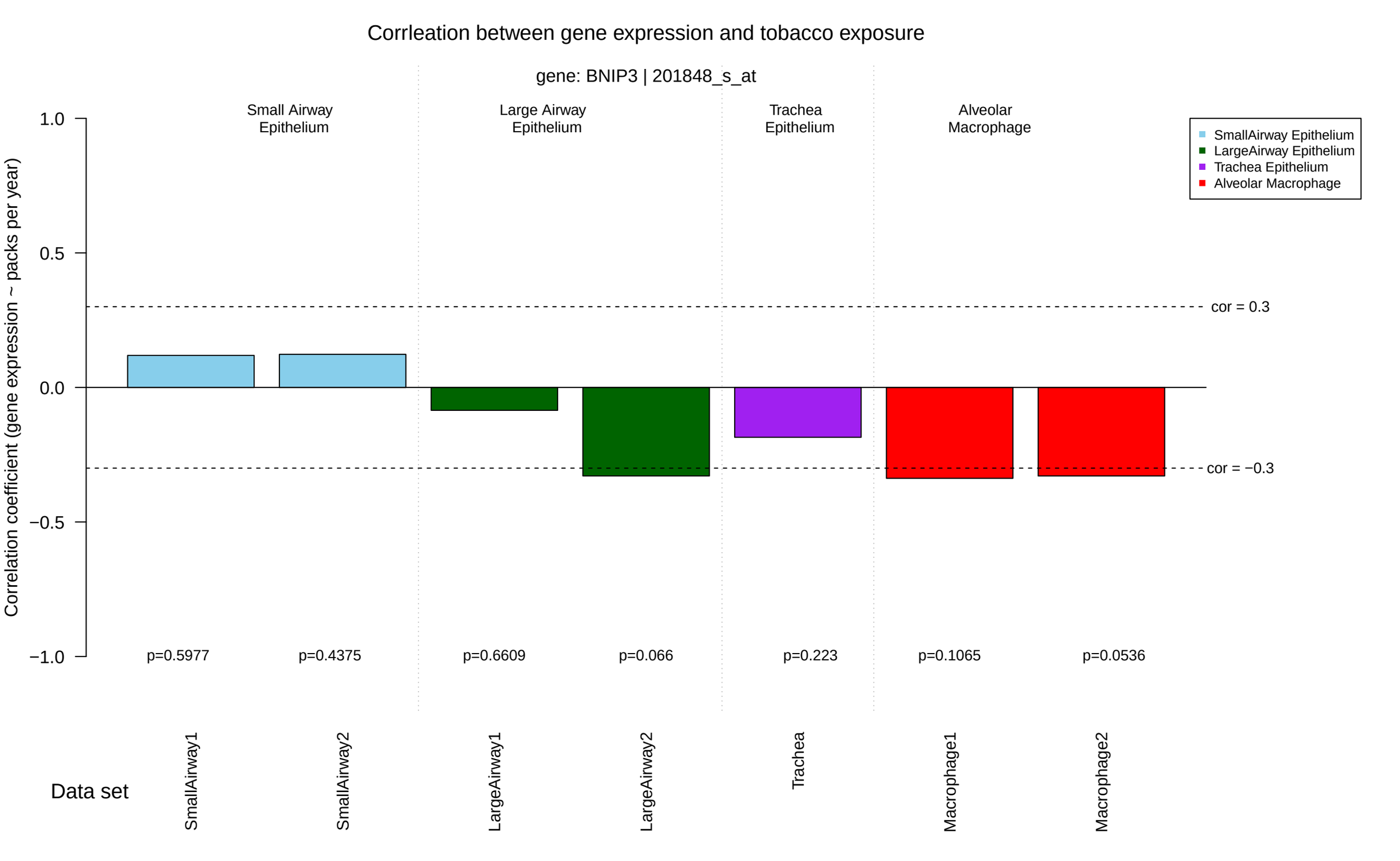

set: LargeAirway1 (p=0.505)
BNIP3 | 201848_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.115)
BNIP3 | 201848_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.018)
BNIP3 | 201848_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4142)
BNIP3 | 201848_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0049)
BNIP3 | 201848_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BNIP3 | 201848_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5977
p=0.4375
p=0.6609
p=0.066
p=0.223
p=0.1065
p=0.0536
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2